

GLOFeagles '26

Automated Detection of Glacial Lakes from Multi-Temporal Satellite Imagery

Team : ICE KATTI

Model Evaluation and Retraining on Round 2 Validation Dataset

To assess the generalization capability of the proposed glacial lake segmentation model, the model trained using the Round 1 dataset was first evaluated on the newly released Round 2 dataset. The Round 2 images were passed through the existing model to generate segmentation masks, and the predicted masks were compared against the provided ground-truth annotations.

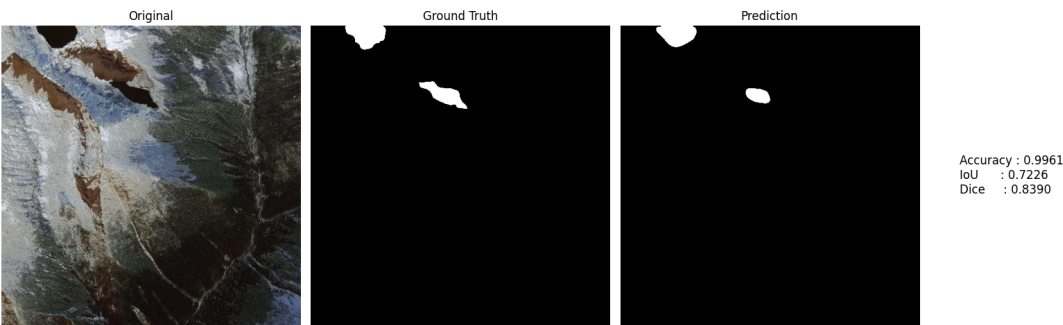
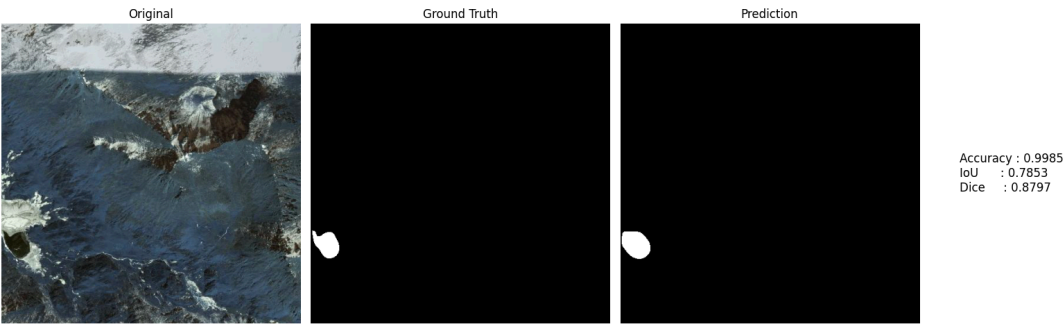
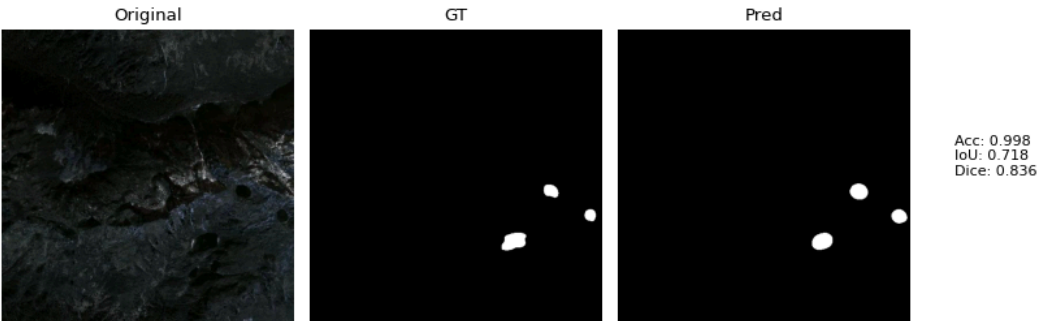
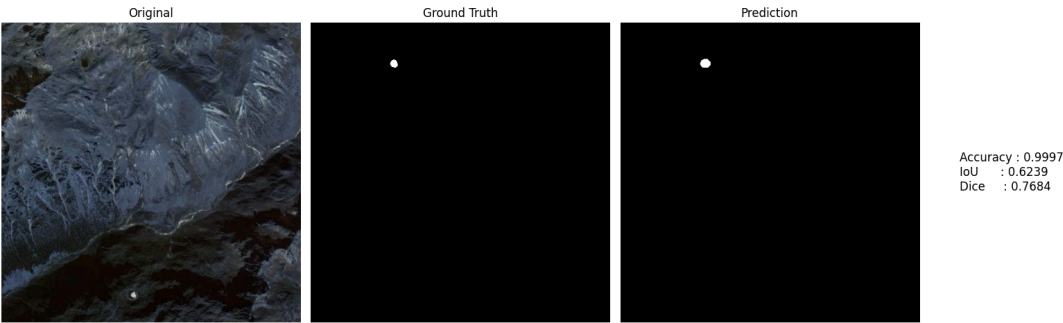
The evaluation revealed that although the model was able to identify major glacial lake regions, its performance was affected by variations in terrain characteristics, lighting conditions, cloud cover, snow cover, debris cover, and water turbidity present in the Round 2 dataset. This domain shift resulted in reduced segmentation accuracy for several categories, indicating the need for additional training on the newly available data.

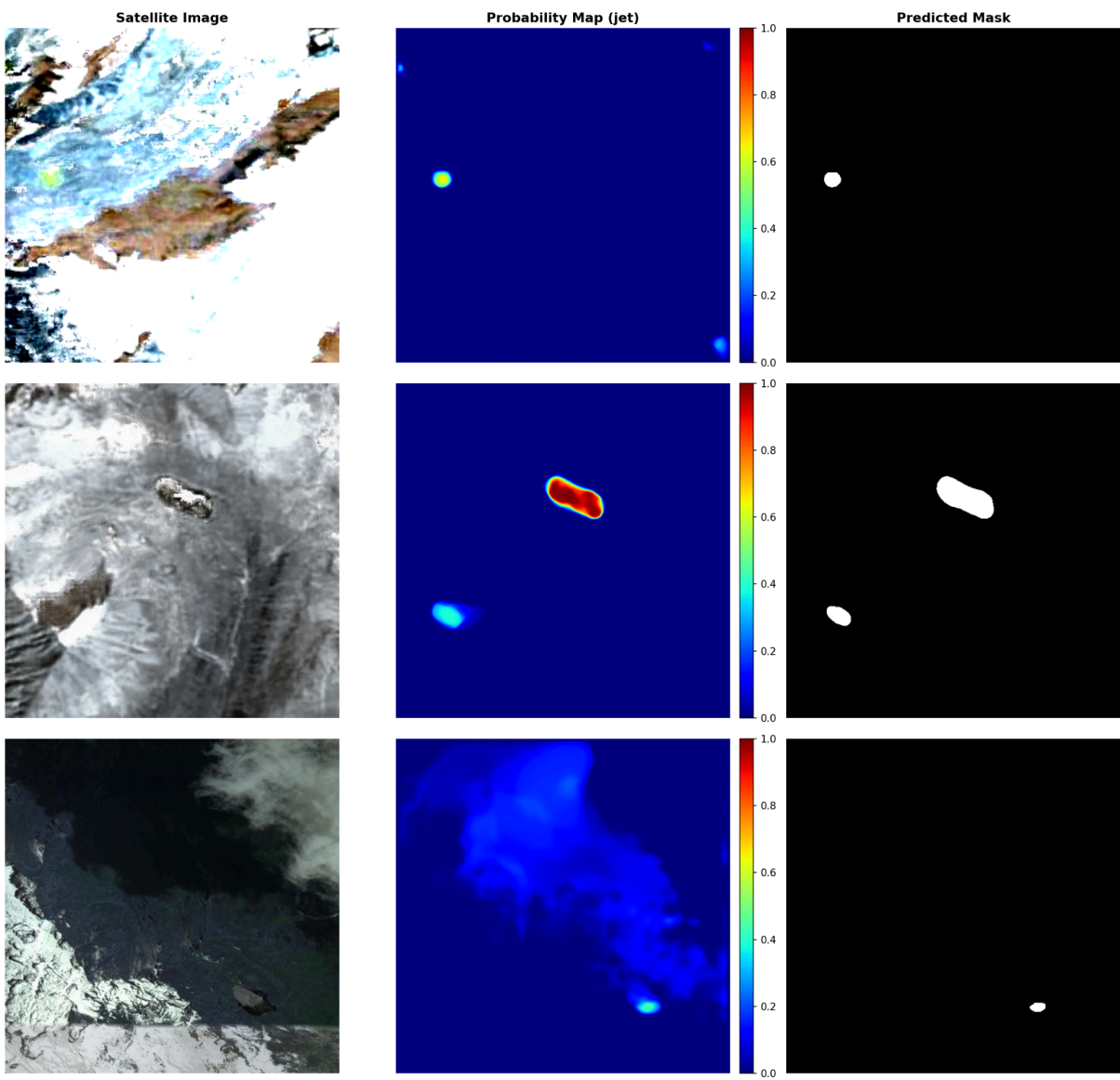
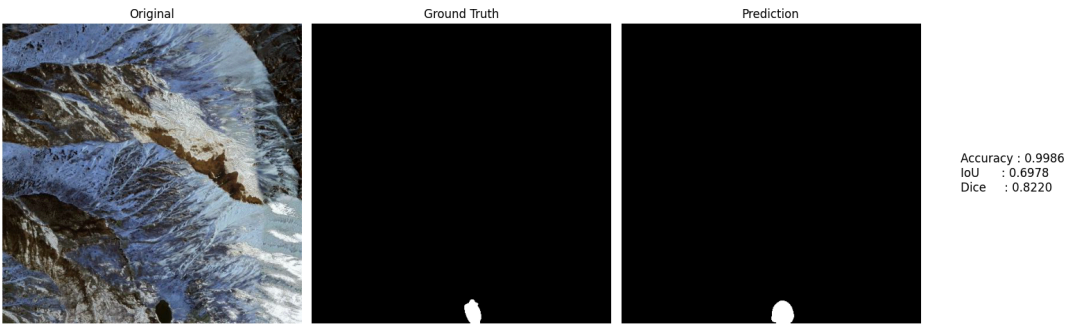
To improve the model's robustness and adaptability, the Round 2 dataset and corresponding annotations were incorporated into the training pipeline. The model was subsequently retrained using the updated dataset while retaining the same segmentation architecture and optimization strategy. This retraining process enabled the model to learn the additional visual patterns and environmental variations represented in the Round 2 imagery.

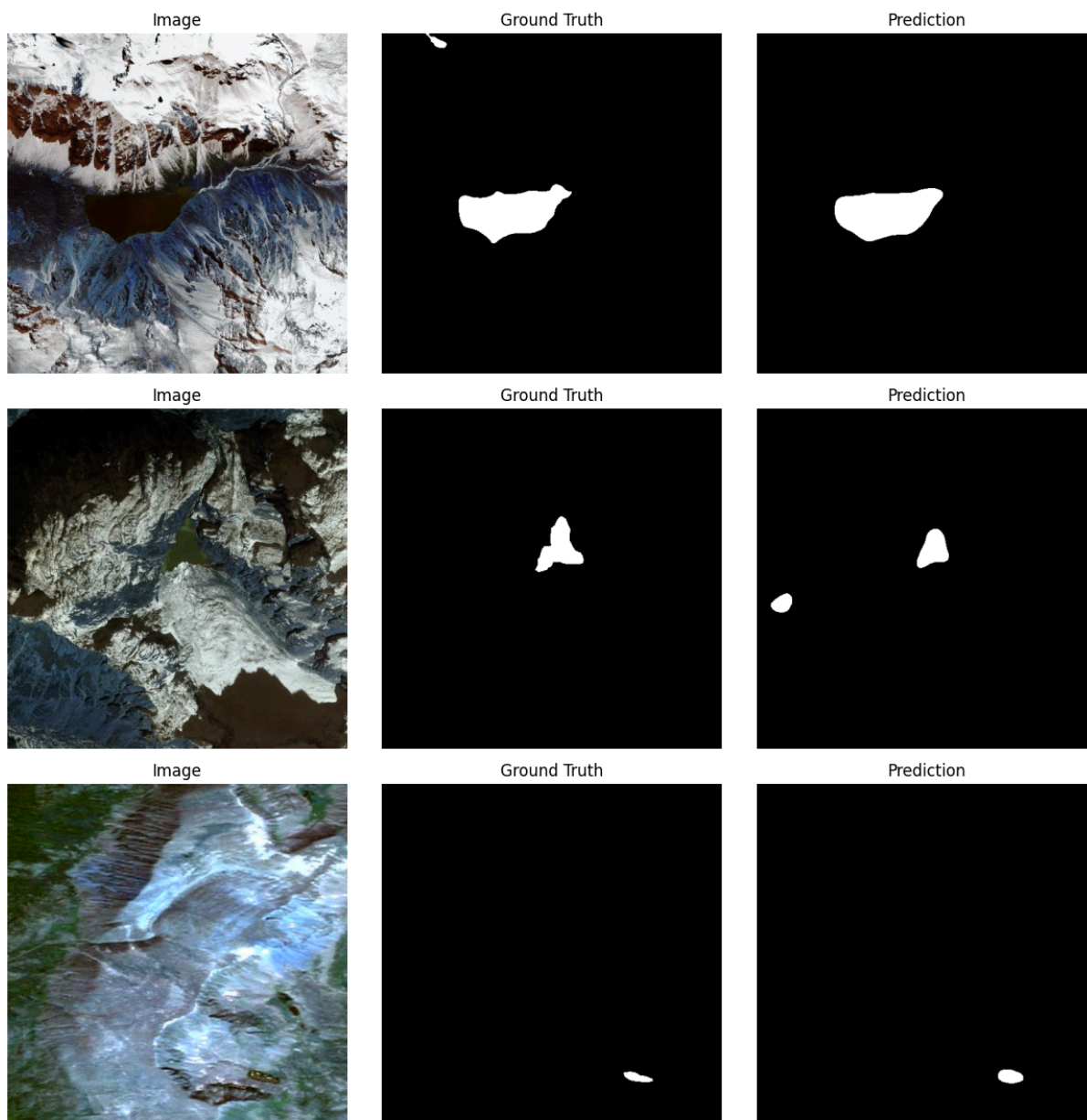
Following retraining, the updated model was evaluated again on the Round 2 dataset. The results demonstrated a significant improvement across all evaluation metrics, including Accuracy, Mean Intersection over Union (IoU), and Dice Coefficient (F1 Score). The retrained model achieved consistently high performance across all six challenge categories, namely Debris Cover, Moraine Dammed Lakes, Terrain Shadow, Varying Turbidity, Snow Cover, and Cloud Cover.

These results confirm that incorporating the Round 2 dataset into the training process substantially enhanced the model's segmentation capability and improved its generalization performance on previously unseen glacial lake imagery.

Ground Truth vs Model Output Comparison

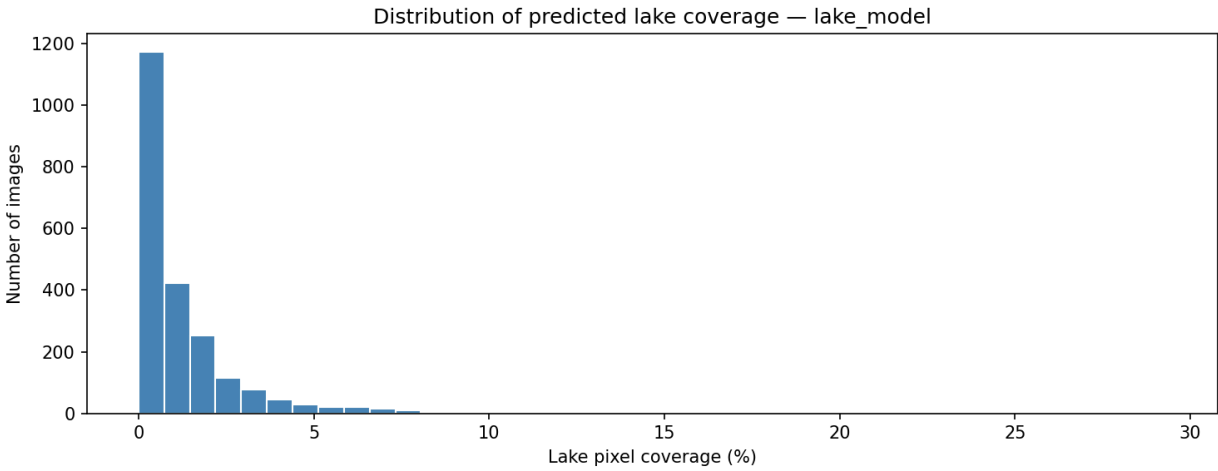






Metric	Value
Mean IoU	0.4211
Min IoU	0.0000
Max IoU	0.9677

Metric	Value
Intersection over Union (IoU)	0.5935
Precision	0.7239
Recall	0.7672
F1 Score	0.7449
Dice Coefficient	0.7449
Specificity	0.9931
Cohen's Kappa	0.7387



Retrained Model Results using round 2 dataset:

Model & Task

- Binary segmentation model to detect lake regions in satellite imagery
- Architecture: U-Net++ with EfficientNet-B3 encoder (ImageNet pretrained)

Dataset

- ~2000+ images across 6 terrain categories: Debris Cover, Terrain Shadow, Snow Cover, Moraine Dammed, Cloud Cover, Varying Turbidity
- Each image paired with a binary ground-truth mask
- Split 70/30 train/validation, stratified so all categories appear in both sets
- Oversampled the Debris Cover category to address class imbalance

Data Augmentation

- Geometric: flips, rotations, shift/scale/rotate, elastic & grid distortion
- Photometric: brightness/contrast jitter, hue/saturation shift, CLAHE, Gaussian noise, gamma shift, random shadows
- Occlusion: coarse dropout
- Goal: generalize across lighting/terrain variation instead of memorizing training data

Training Setup

- Loss: combined Dice + Focal + boundary-aware loss (extra penalty near lake edges)
- Optimizer: AdamW with cosine-annealing learning rate schedule
- Validation tracked via IoU and Dice every epoch

Post-Training Tuning

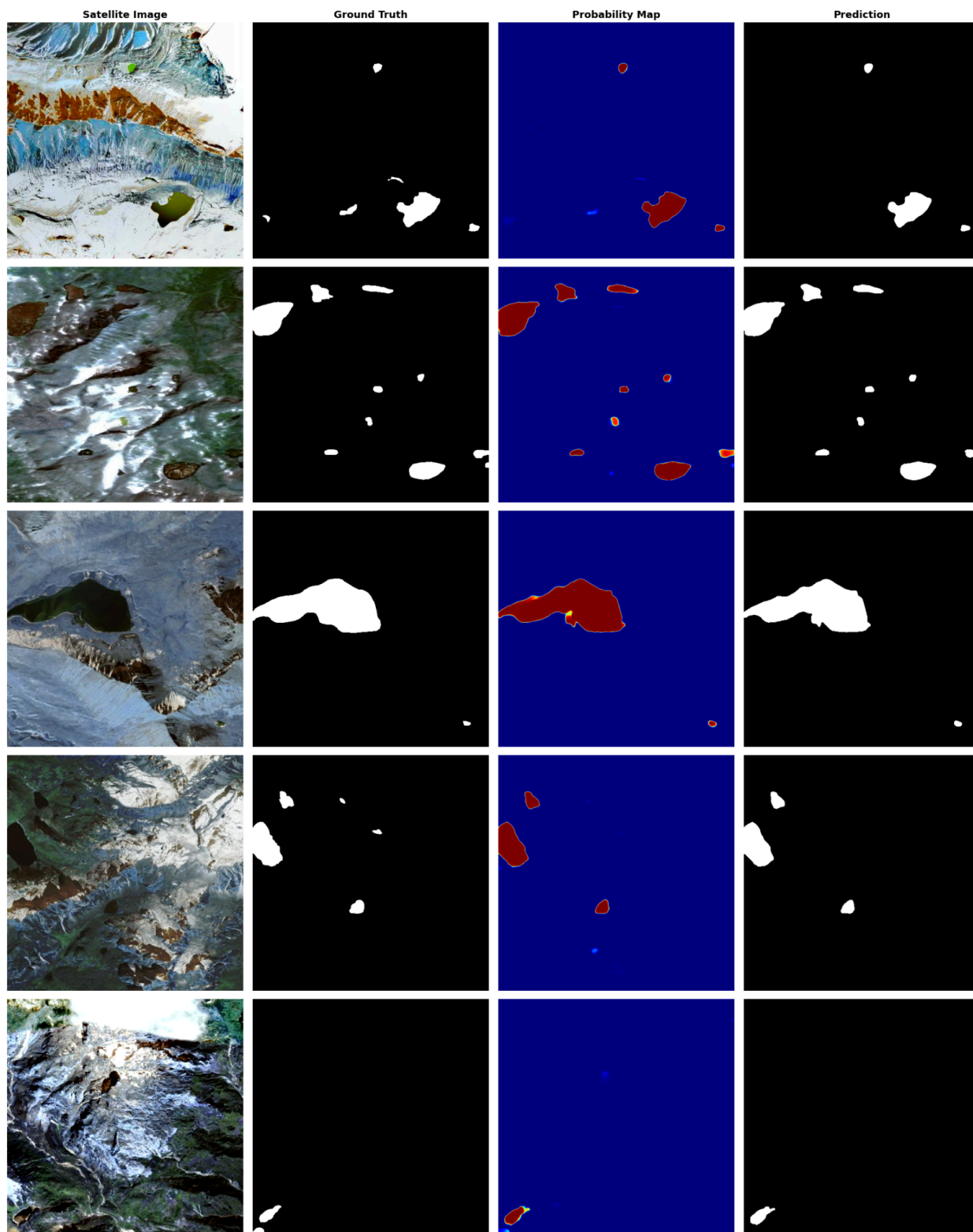
- Ran a threshold sweep on validation set to find the cutoff that maximized IoU
- Saved both raw model weights and a full checkpoint (weights + tuned threshold + metrics) to Drive

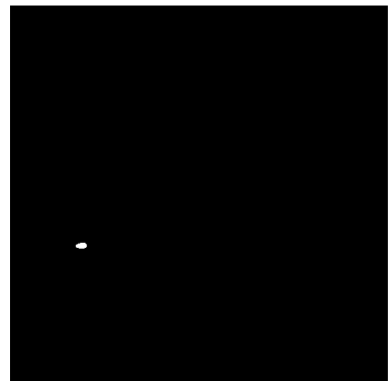
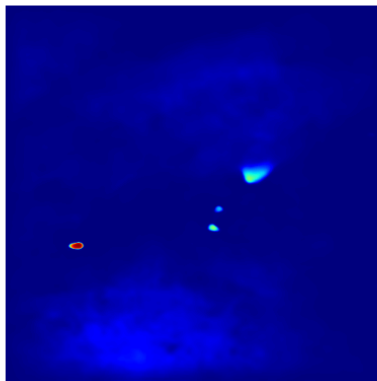
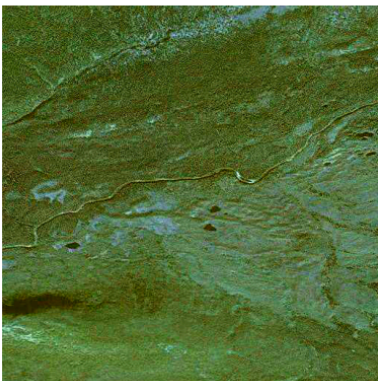
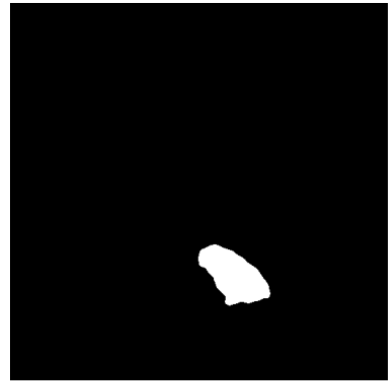
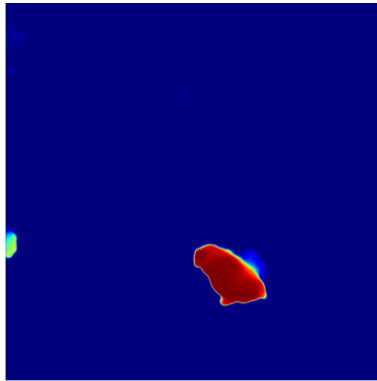
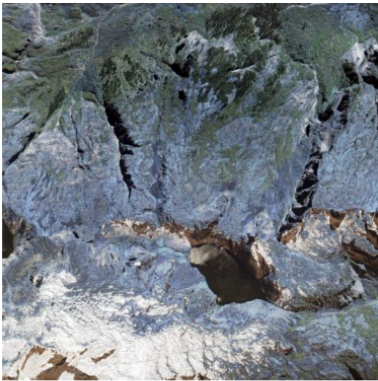
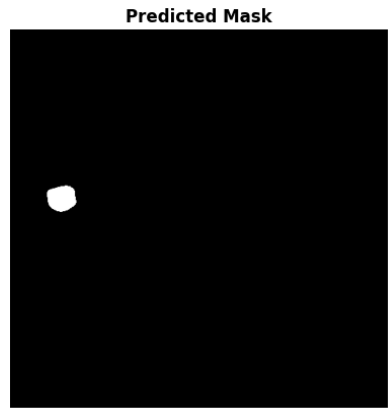
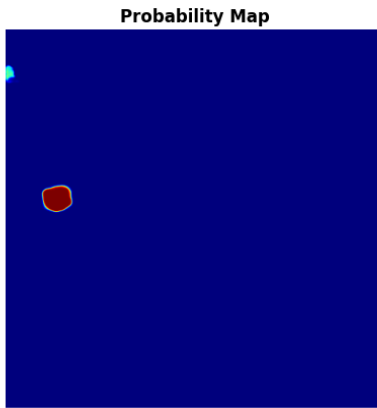
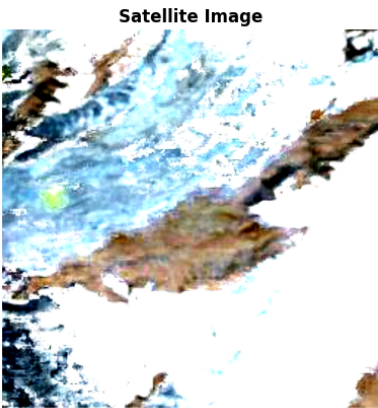
Inference Pipeline (making it "work")

- Built a test-time augmentation (TTA) function: runs each image through the model at multiple scales + multiple flip/rotation variants
- Averages resulting probability maps, applies the tuned threshold
- Cleans up final mask using morphological operations + small-blob removal

Reuse / Validation

- Built a separate inference-only notebook that loads the saved checkpoint (no retraining)
- Reused the same TTA function to generate masks on a new validation set
- Organized output masks by category, matching source folder structure
- Computed per-image Accuracy/IoU/Dice against ground truth to confirm consistent performance outside the original training run





Class Type	Accuracy	Mean IoU	F1/Dice
Debris Cover	0.9958	0.7029	0.7768
Moraine Dammed	0.9905	0.6673	0.7308
Terrain Shadow	0.9956	0.6815	0.7540
Varying Turbidity	0.9949	0.6941	0.7687
Snow Cover	0.9960	0.6562	0.7238
Cloud Cover	0.9990	0.8698	0.9280
Average Metric Values	0.9953	0.7119	0.7804